

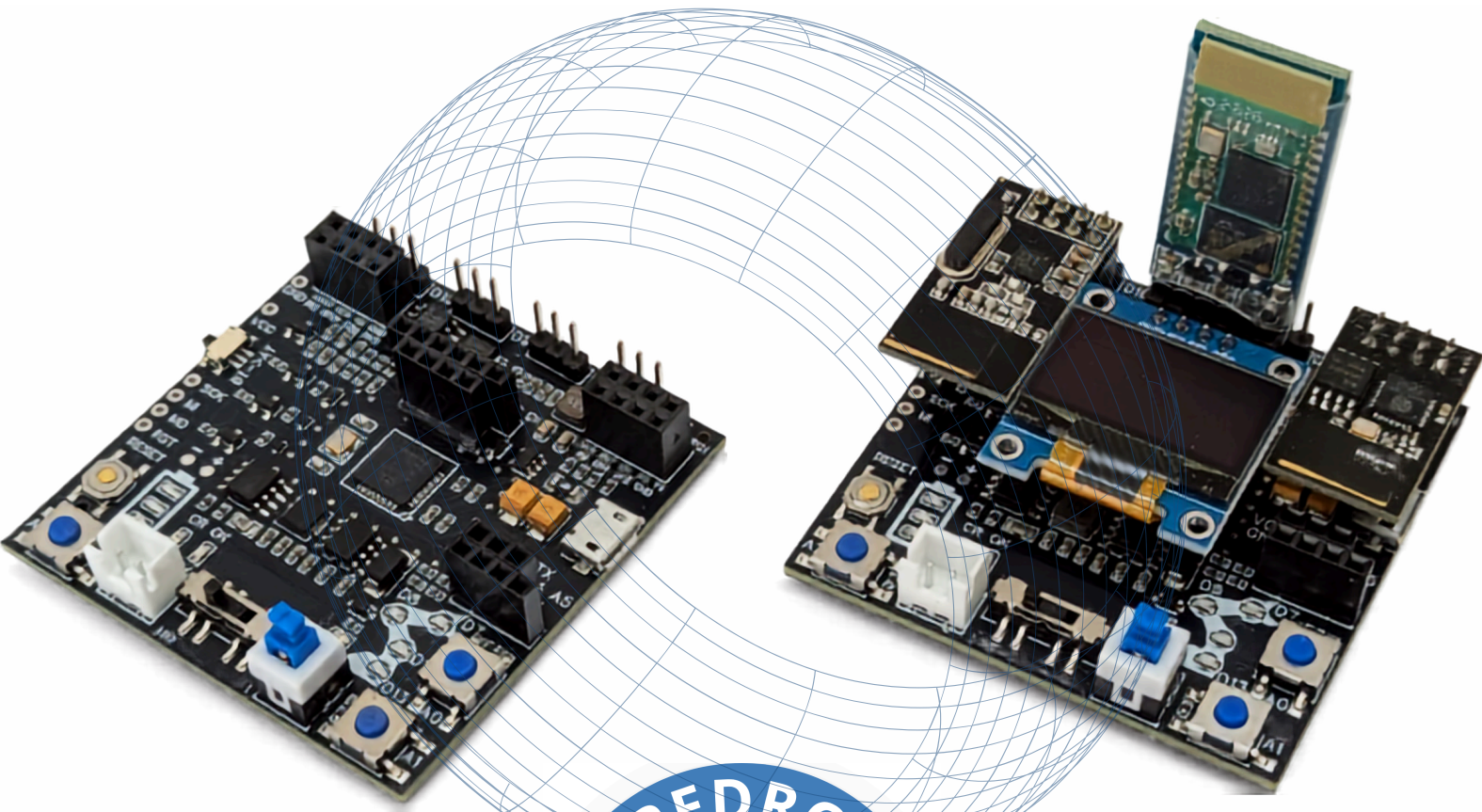
BEGINNER

pedro



STEM Program

Lesson #2: Pedro Board



Complete this lesson
to earn your Pedro
Badge



Add it to your Pedro
Passport to track
your STEM journey

YOUR MISSION: Discover the Pedro Board, the electronic brain that controls Pedro robot.

Learning Goals

In this lesson, students will discover the **Pedro Board**, the electronic brain that controls the Pedro robot. They will explore how **hardware and software interact** in an embedded system and learn how electronic board control a robot.

Students will also upload their **first Arduino program** and interact with the robot's components.

At the end of this lesson, you will unlock your Pedro Badge.

 the Pedro Board Badge



Teacher Notes

- No prior robotics experience required.
- Duration: 60–90 minutes
- Difficulty: ● Beginner
- Required Materials
 - **1 Pedro control board with USB cable per student or per group**
(recommended: groups of 2 students)
 - **1 computer** with the Arduino IDE installed



Think: The **brain icon** encourages students to make careful observations, share their ideas, and engage in discussion by explaining and justifying their answers.



Learn: The **light bulb icon** indicates important concepts that students should read carefully. Students are encouraged to ask questions and discuss these topics about the robotics concepts presented in the lesson.



Make: The **pencil icon** indicates that students must complete the activity on the page. These activities are designed to reinforce the concepts learned through experimentation, and problem-solving.



Embedded System

An **embedded system** is a small computer designed to perform one or more specific tasks inside a device.

It combines **electronics** and **software** to control hardware automatically.

When an embedded system is connected to a mechanical structure, it becomes a **robotic system** capable of sensing, thinking, and acting.

In robotics, three engineering disciplines work together:

- ⚡ **Electronics** – controls the electrical components.
- 💻 **Programming (Software)** – tells the robot what to do.
- ⚙️ **Mechanics** – provides movement and physical structure.



Embedded systems are everywhere in our daily lives. Examples include:

- ✈️ Aircraft and drones
- 🚗 Cars and electric vehicles
- 📱 Smartphones
- 🚀 Rockets and satellites

Most modern electronic devices contain one or more embedded systems.



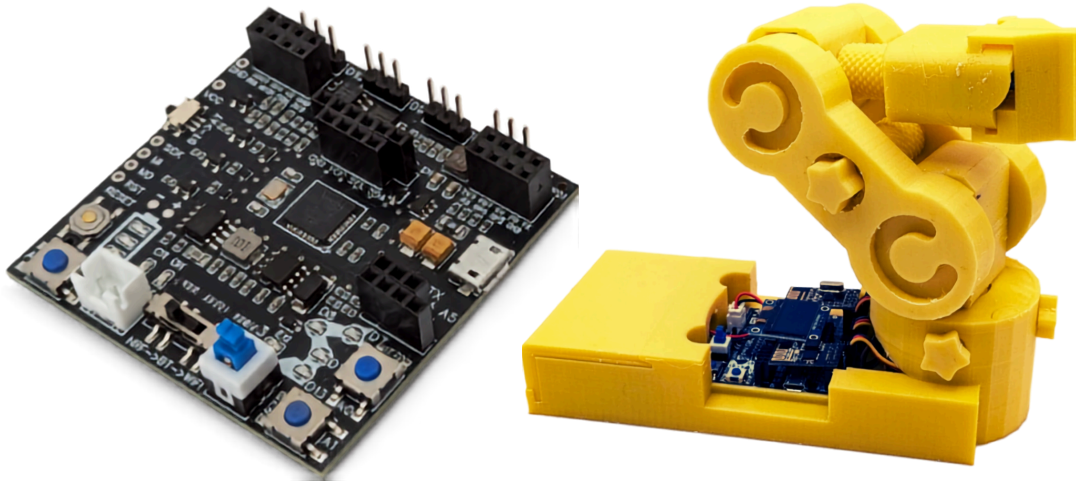
Can you name some everyday devices that contain an embedded system?



Pedro as an Embedded System

Pedro is an example of an **embedded system**.

Without **electronics**, **software**, or **mechanics**, Pedro would not be able to operate.



The most important component on the board is the **microcontroller**. In robotics, the microcontroller can be considered the **brain of the robot** because it executes the program, processes information, and controls all of Pedro's movements. It's a tiny computer built onto an electronic board.

Pedro is powered by an **ATmega32U4** microcontroller. It reads the program stored in its memory, receives data from communication interfaces, and sends commands to the servomotors to control the robot's actions.

Why does Pedro need both electronics and software to move?
What would happen if Pedro's electronic board stopped working?
Which part of an embedded system acts like the robot's brain?
Why?



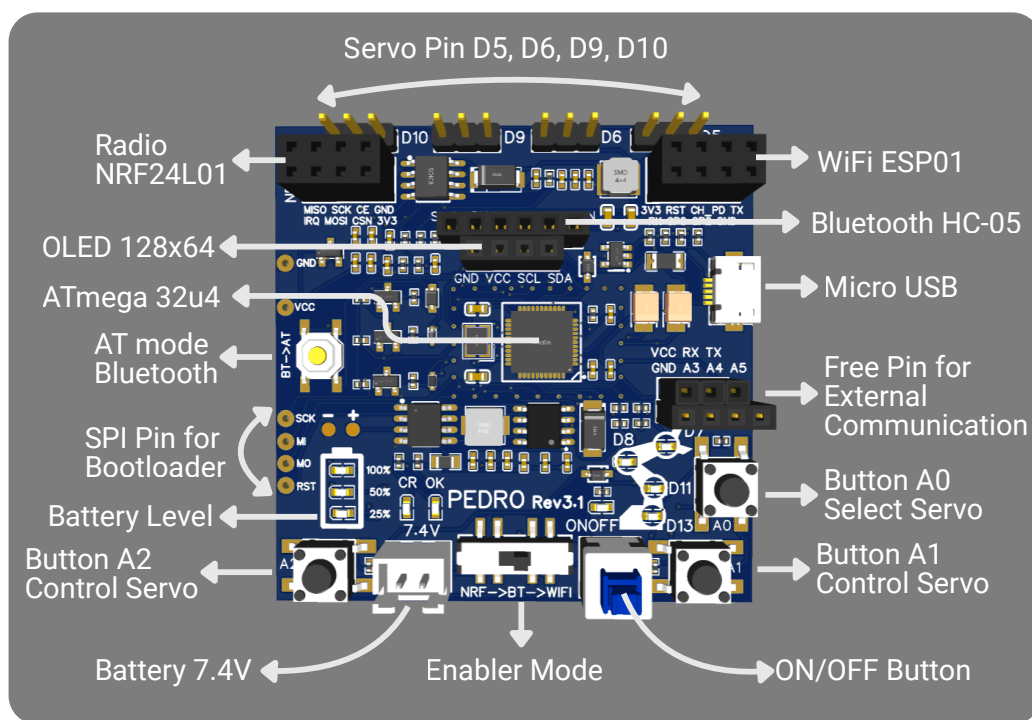
Encourage students to realize that embedded systems are found everywhere not only in robots. Explain that almost every modern smart device uses a small computer to control its functions. Pedro is a practical example that combines **mechanics**, **electronics**, and **programming** into a single STEM learning platform.



Pedro Board Features

This board contains all the electronic components needed to control the robot, including its power management, connectors, and communication interfaces.

The battery supplies energy to the entire system, while the software controls every movement. The Pedro control board is fully **compatible with the Arduino development environment**, allowing students to program the robot using the **Arduino IDE** and the **C/C++ programming language**.



The **USB port** has **three main functions**:

1. **Programming** the board
2. **Charging** the battery
3. **Controlling** the Pedro robot via a **graphical interface**

Why does Pedro need both electronics and software to move?



Encourage students to realize that embedded systems are found everywhere not only in robots. Explain that almost every modern smart device uses a small computer to control its functions. Pedro is a practical example that combines **mechanics, electronics, and programming** into a single STEM learning platform.



Memory Architecture of the Pedro Robot

The Pedro robot uses an **ATmega32U4 microcontroller** as its main processing unit. This microcontroller contains several types of memory, each designed for a specific purpose.

The ATmega32U4 has the following memory resources:

Memory Type	Size	Main Function
Flash Memory	32 KB	Stores the compiled robot program (firmware, will be introduced in Lesson #4)
SRAM (RAM)	2.5 KB	Stores temporary data while the robot is running (will be introduced in Lesson #7)
EEPROM Memory	1 KB	Stores permanent user data and recorded sequences

A small robot such as Pedro has only a limited amount of memory compared to a computer.

For example:

- A smartphone can store **hundreds of gigabytes** of data.
- The ATmega32U4 has only **1 KB of EEPROM** for saved sequences.

This limitation requires engineers to carefully decide:

- What information must be stored?
- How can the data be represented using fewer bytes?
- How can the program be optimized?



Why does the ATmega32U4 need different types of memory instead of using only one memory type?



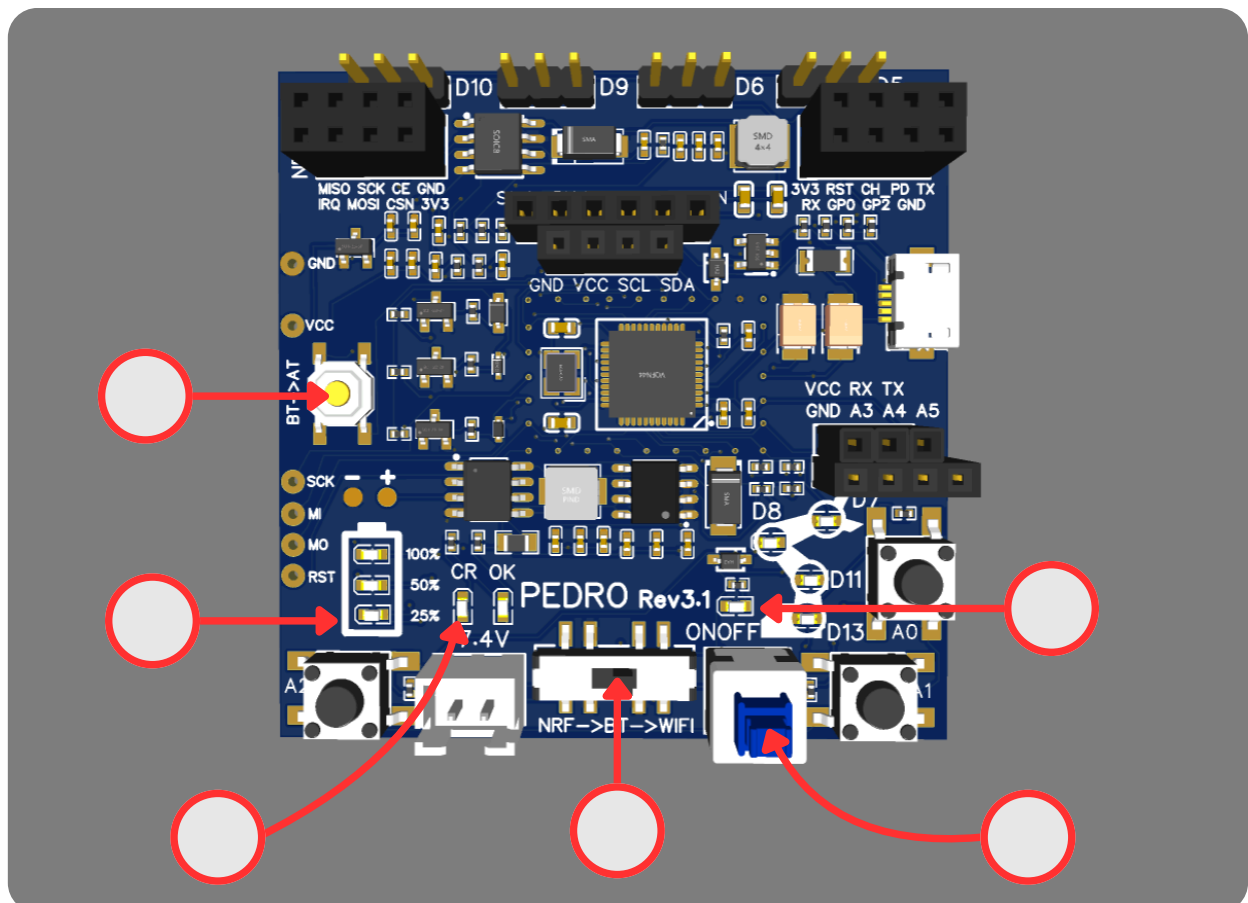
Identifying Board Components

Students work in small groups and try to identify the elements on the board by adding the number element

Part 1: Identifying Non-Programmable Components

Non-programmable components are electronic parts that provide information or allow user interaction without requiring any programming. Examples include **LEDs, switches, and push buttons**.

1- Button ON/OFF	2- LEDs ON/OFF	3- Battery Level LED
4- LED CR OK	5- Enabler Bluetooth AT mode	6- Switch mode: radio, BT, WiFi

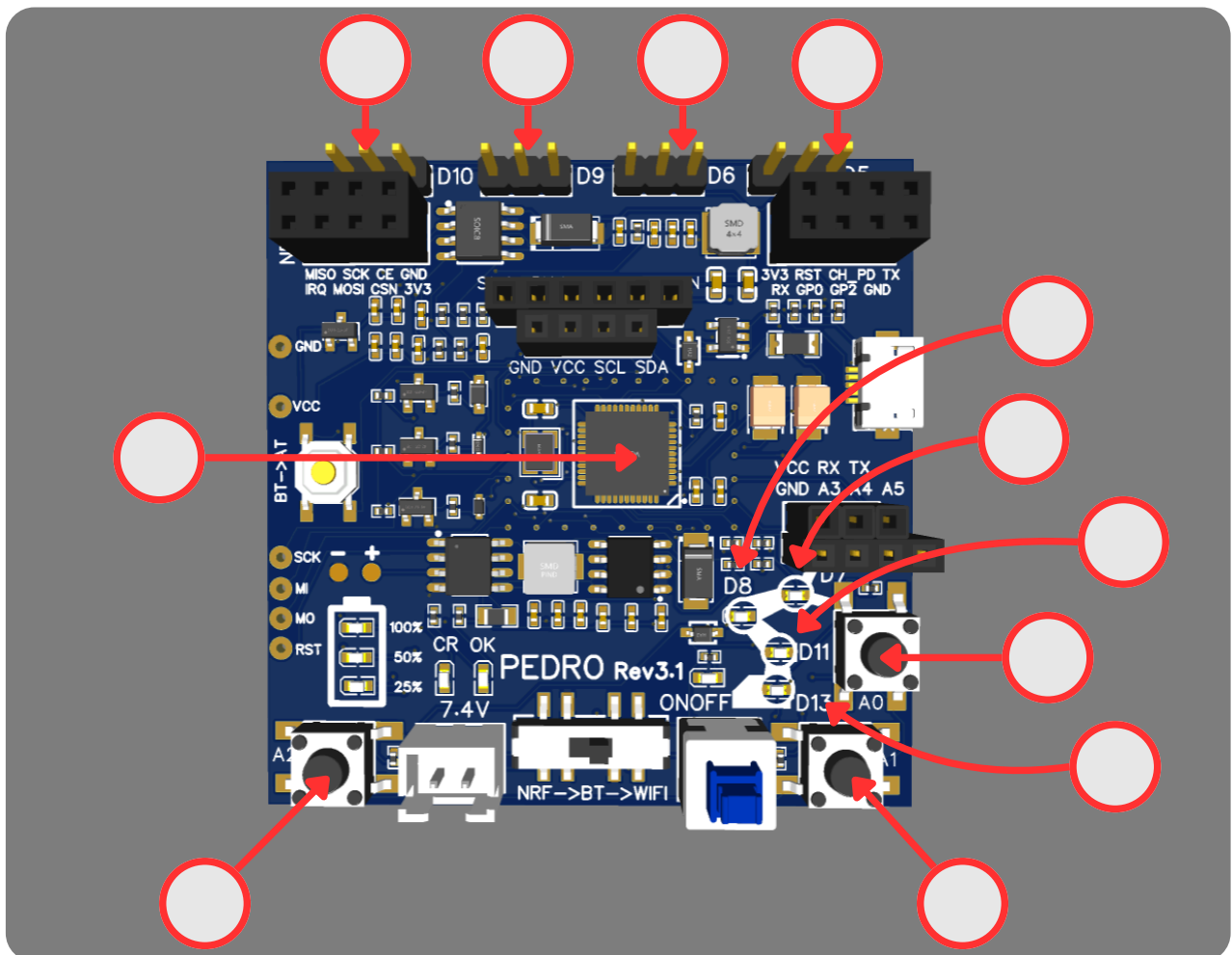


- LED CR: Indicates that the battery is currently charging.
- LED OK: Indicates that the battery is fully charged.
- Battery Level LEDs: Show the battery charge level from bottom to top (25%, 50%, and 100%).

Part 2: Identifying of Programmable Components

A **programmable component** is a component on the electronic board that can be controlled by programming the **microcontroller**. This can include **buttons**, **LEDs**, or **input/output pins**.

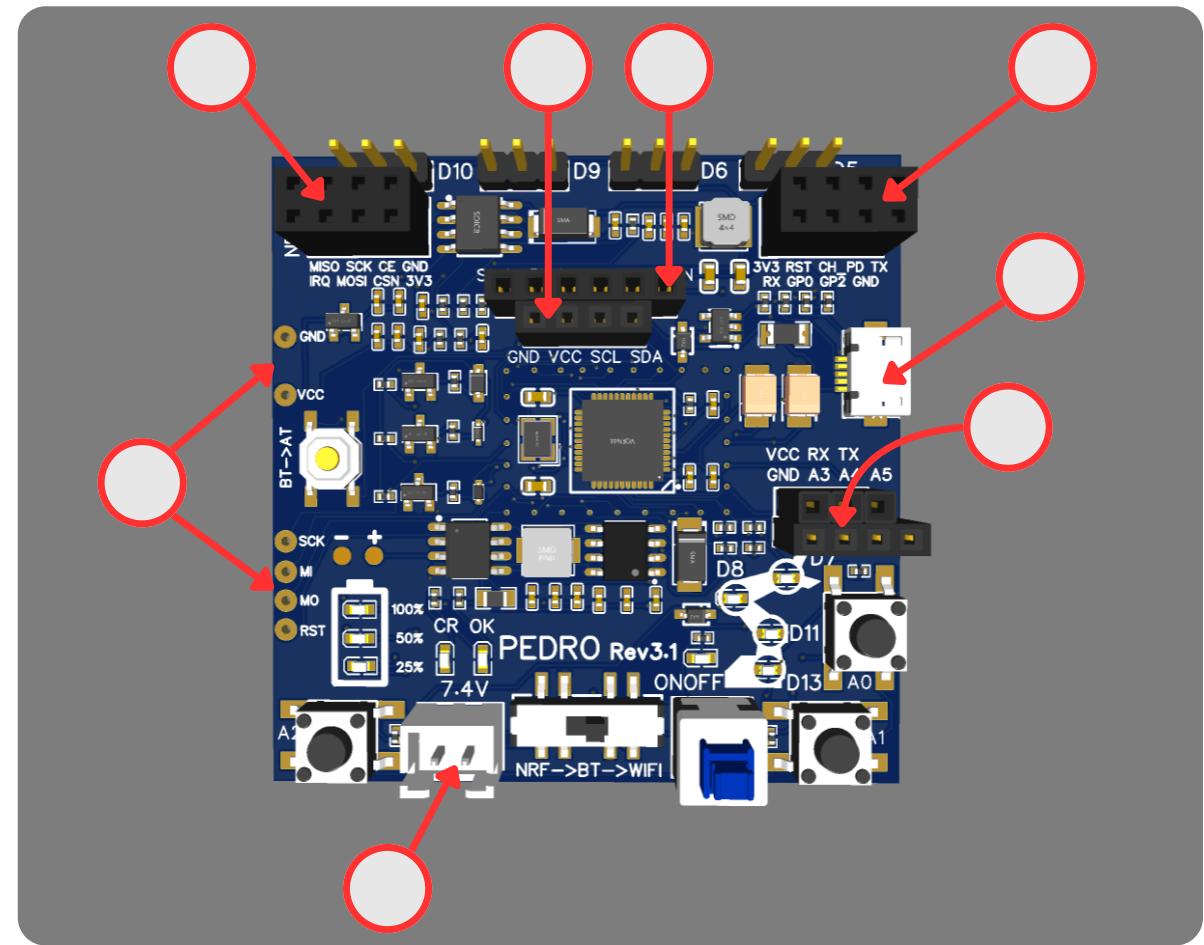
1- LED Base D13	2- LED Shoulder D11	3- LED Elbow D8	4- LED Gripper D7
5- Pin Servo Base D5	6- Pin Servo Shoulder D6	7- Pin Servo Elbow D9	8- Pin Servo Gripper D10
9- Button A0 Select Servo	10- Button A1 Control Servo	11- Button A2 Control Servo	12- ATmega32U4 Microcontroller (the brain)



Part 3: Identifying Connectors

Connectors allow the electronic board to communicate with other devices or connect external components such as **sensors, servomotors, batteries, and computers.**

1- Radio module NRF24L01	2- WiFi module ESP01	3- Bluetooth module HC-05	4- OLED Screen 128x64
5- 7.4V Battery Connector	6- Free Pin	7- Micro USB port	8- SPI Bootloader Pin



 What are the three main functions of the Micro USB port?



Arduino IDE

The **Arduino IDE (Integrated Development Environment)** is the software used to write, compile, and upload programs to the Pedro control board.



Using the Arduino IDE, students can create programs that control LEDs, read buttons and sensors, and move servomotors. Programming allows Pedro to perform actions automatically by following a sequence of instructions written by the user.

Programming a robot follows four simple steps:

1. **Write the Program:** Write the code in the Arduino IDE. (based on C/C++).
2. **Verify (Compile) the Program:** the Arduino IDE checks the code for errors.
3. **Upload the Program:** Connect Pedro with a Micro USB cable and click **Upload**. The program is copied to the **ATmega32U4** microcontroller.
4. **Run the Program:** Pedro Board starts running the program automatically.

Every robot receives information (**inputs**) and performs actions (**outputs**).

- ☐ **Inputs** provide information to the microcontroller. The microcontroller **reads** these inputs before making a decision. Examples on Pedro: Push buttons, Bluetooth commands, Radio commands
- ☐ **Outputs** allow the robot to interact with its environment. The microcontroller **controls** these outputs according to the program. Examples on Pedro: LEDs, Servomotors



Tell students that **the program is not executed on the computer**. When they click **Upload**, the Arduino IDE copies the program into the **ATmega32U4 microcontroller**, which acts as Pedro's brain. Once the upload is complete, Pedro no longer needs the computer to execute the program.

- **Download and install** the latest version of the Arduino IDE:
<https://www.arduino.cc/en/software>



Programmable Components

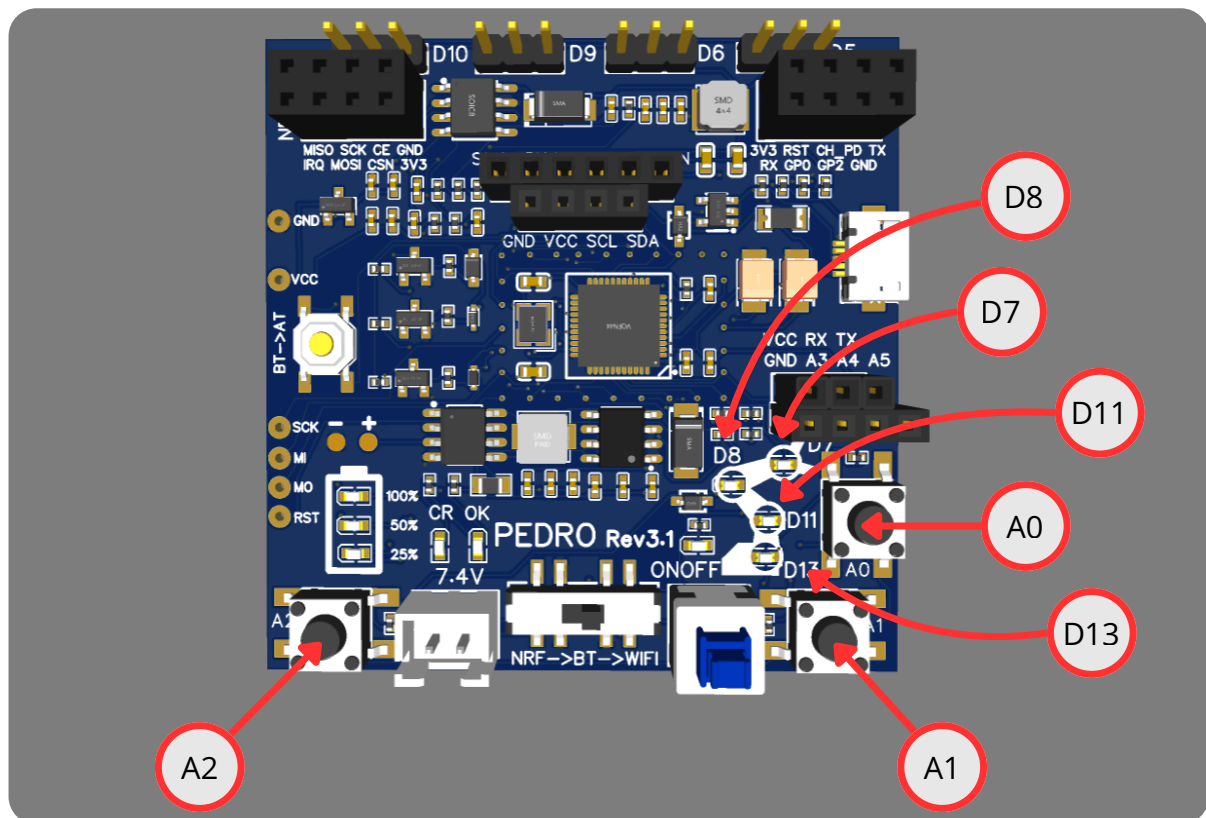
Students will interact with the following components:

Digital Outputs

- LED D13
- LED D11
- LED D8
- LED D7

Digital Inputs

- Button A0
- Button A1
- Button A2



Key Vocabulary

- **Program** → A set of instructions for the robot.
- **Code** → The text written by the programmer.
- **Upload** → Transfer a program from the computer to the microcontroller.
- **Compile** → Convert the code into machine language.
- **Microcontroller** → The robot's brain that executes the program.
- **Input** → Information received by the robot.
- **Output** → An action performed by the robot.



Your First Program

1- Blink LED

- **Connect** the Pedro board to your computer with the Micro USB cable.
- Open **Arduino IDE**
- Select the port that appear when you connect Pedro robot: **Tools → Port**
- Select the board: **Tools → Board → Arduino AVR Boards → Arduino Micro**
- Open the Blink example: **File → Examples → 01.Basics → Blink**
- You should see a program with **two functions**:
 - **setup()**: Runs **once**. It initializes the **LED D13** as an **output**.
 - **loop()**: Runs **continuously**.

```
// the setup function runs once when you press reset or power the board
void setup() {
  pinMode(13, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(13, HIGH);
  delay(1000);
  digitalWrite(13, LOW);
  delay(1000);
}
```

- **Compile and upload** the program to your Pedro board.

Experiment:

1. Which LED turns ON?
2. Does it stay ON or blink?
3. Change **HIGH** to **LOW** then upload the program
4. What happens ?
5. Restore the original code and upload it again.
6. Change delay(**1000**) to delay(**500**) then upload the program.
7. Is the blinking faster or slower?
8. Try other delay values such as: **100, 2000, 5000** and see what happens.
9. Change the LED pin number then upload the program again and observe what happens. Try these pins: **7, 8, 11, 14**

Coding Challenge:

Modify the program so that all four **LEDs (D7, D8, D11, and D13)** blink at the same time.



Your Second Program

2- Button

- Now open the Button example: **File → Examples → 02.Digital → Button**
- You should see a program with **two functions**:
 - **setup()**: Runs **once** when Pedro starts. It initializes a **ledPin** as an **output** and a **buttonPin** as an **input**
 - **loop()**: Runs **continuously**.
- Set **buttonPin** value to **A1** and **ledPin** to **8**
- **Compile and upload** the sketch to your Pedro board.

Experiment:

1. Press the **button A1** onto the Pedro board.
2. What happens ?
3. Change the **ledPin** value by **7, 11, 13** and observe what happens when you press the **button A1**. Don't forget to upload program after each change.
4. Now change the **buttonPin** value by **A0**?
5. What happens when you press again on the **button A1**? why?
.....
6. Press the **button A0**. What happens ?
.....
7. Now change the **buttonPin** value by **A2**?
8. What happens when you press again on the **button A0**? why?
.....
9. Press the **button A2**. What happens ?
.....

Coding Challenge:

Modify the program so that **all four LEDs (D7, D8, D11, and D13)** turn **ON at the same time** whenever **button A0, A1, or A2** is pressed.

Your program should:

- Read the three buttons.
- Check whether **at least one button** is pressed.
- Turn **all four LEDs ON**.
- Otherwise, turn **all four LEDs OFF**.



Quiz - Check Your Understanding

1. What type of microcontroller powers Pedro?

- ☐ ATmega32U4
- ☐ ESP32
- ☐ Raspberry Pi
- ☐ STM32

2. What software is required to program Pedro?

- ☐ Visual Studio
- ☐ Arduino IDE
- ☐ Python
- ☐ MATLAB

3. What component allows Pedro to communicate wirelessly?

- ☐ NRF24L01 module
- ☐ HC-05 module
- ☐ GPS module
- ☐ Ultrasonic sensor

4. What is the screen resolution of Pedro's OLED display?

- ☐ 64x64
- ☐ 128x32
- ☐ 128x64
- ☐ 256x128

5. Which component receives and processes the user's commands?

- ☐ The servo motors
- ☐ The microcontroller
- ☐ The battery
- ☐ The Bluetooth module

6. Which component represents the “brain” of the Pedro board?

- ☐ The OLED Display
- ☐ The ATmega32u4 Microcontroller
- ☐ The Battery Charger
- ☐ The NRF24L01 Radio Module

7. What type of display is integrated into the Pedro Board?

- ☐ LCD
- ☐ OLED
- ☐ TFT
- ☐ LED

8. What function does the Micro USB port serve?

- ☐ Power only
- ☐ Programming, charging, and serial control
- ☐ Bluetooth pairing
- ☐ WiFi connection

9. Which module provides WiFi capability?

- ☐ NRF24L01
- ☐ HC-05
- ☐ ESP8266-01
- ☐ GPS module

10. What does the ATmega32u4 allow students to do?

- ☐ Design 3D parts
- ☐ Program the robot using Arduino IDE
- ☐ Print STL files
- ☐ Connect motors directly

11. Which component shows the battery charge level?

- ☐ LED CR
- ☐ LED OK
- ☐ Battery Level LEDs
- ☐ OLED screen

12. What does the LED “CR” indicate?

- ☐ Bluetooth connected
- ☐ Battery charging
- ☐ Low voltage
- ☐ Power OFF

Congratulations Mission Completed!

You have unlocked:  the Pedro Board badge
Don't forget to stick your Pedro Badge into your Pedro Passport before starting the next lesson.

